WAN MUHAMMAD ASHRAF WAN YUSOFF

Senior NPI Technical Lead | Semiconductor Final Test Engineering
Kuala Lumpur / Klang Valley, Malaysia | +60 14 823 9695 | wanashraf2127@gmail.com | linkedin.com/in/wanashraf

PROFESSIONAL SUMMARY

NPI technical lead with 5 years of engineering experience, including 4+ years in semiconductor Final Test NPI at NXP Semiconductors. Owned and delivered 50+ New Product/Technology Introduction (NPI/NTI) hardware projects end to end with 100% First Time Right (FTR) and 100% On-Time Delivery (OTD), zero design re-spin and a 20% cycle-time reduction. Combines project ownership (scope, schedule, budget, risk, suppliers, global stakeholders) with deep technical leadership in test hardware, PCB design, DOE and CFD/FEA. Trusted on revenue-critical programs, including a platform conversion that secured ~US$40M. First Class Mechanical Engineering graduate; Lean Six Sigma Green Belt trained; certified in machine learning and data science.

CORE COMPETENCIES

Program Leadership: NPI/NTI lifecycle • Scheduling & milestone tracking • Budget & cost control • Risk management • Global stakeholder & supplier management • Production release readiness

Technical Depth: Final Test hardware (loadboards, sockets, handler interfaces) • PCB design • DOE • CFD / FEA • DFMEA, 3x5 Why, MSA, SPC • Lean Six Sigma • Data analysis (Power BI, JMP, Minitab) and AI/ML

KEY ACHIEVEMENTS

- **100% FTR and 100% OTD** across 50+ NPI/NTI hardware projects, with zero design re-spin and a 20% cycle-time reduction (shift-left).
- **~US$40M revenue secured** by leading the Teradyne J973 to UltraFlex UP1600 test platform conversion for the MPC8308 product line on an aggressive schedule, while avoiding NRE cost through in-house loadboard development.
- **3x contact pin life (300k to 1M touchdowns)** from cantilever pins introduced on MCT SH-5300 and FH-1200 handlers, cutting downtime and improving MTBF; rolled out across multiple packages and the new Cohu Jaguar in-strip handler.
- **Top Achiever FY2023 and FY2024 at NXP;** four recognition awards (SPOT and Technical Excellence), 2023 to 2025.
- **RM128,000 annual cost saving** at Top Glove from two automation and digitalisation projects.

PROFESSIONAL EXPERIENCE

NPI Hardware Design Engineer / Project Owner | NXP Semiconductors **Sept 2022 – Present**

NXP Semiconductors Sdn. Bhd., Petaling Jaya, Selangor | Multinational semiconductor company | Final Test NPI/NTI for automotive and industrial devices

- Own the full NPI/NTI hardware lifecycle for 50+ projects (requirements, design, supplier procurement, build, qualification, production release), delivering 100% FTR and 100% OTD.
- Plan and track schedule, budget and risk on every project, using Excel-based project plans with automated Gantt charts; cut project cycle time by 20% and met shift-left goals.
- Lead global coordination with NXP teams in the USA, Europe and Asia and with multiple external suppliers, resolving technical issues to keep milestones on track with zero design re-spin.
- Led the J973 to UP1600 conversion (~US$40M revenue secured) and the cantilever pin introduction on MCT SH-5300/FH-1200 handlers (pin life 300k to 1M touchdowns), extended to VSON, SO, HVQFN and DHVQFN packages and the Cohu Jaguar handler.
- Design PCB layouts and test hardware (sockets, pogo pins, handler conversion kits, tester-handler docking) on Teradyne (UltraFlexPlus, UltraFlex UP1600, J750) and Advantest (Exascale Duo, PinScale PS1600) testers and Delta, MCT, Esmo Talos, Seiko Epson and Cohu Jaguar handlers.
- Run DOE to validate hardware reliability and optimise insertion count; run CFD and FEA simulation before production release; maintain drawings and specifications for procurement and manufacturing readiness.

Production Engineer | Top Glove Corporation Berhad **Oct 2021 – Sept 2022**

Meru, Selangor | Top Performer (Midterm Grading) FY2022

- Delivered machine design and automation projects worth RM128,000 in combined annual savings, bridging design and manufacturing.

- Led Continuous Improvement Projects (CIP) and digitalised packaging and monitoring processes with Power BI (Industry 4.0).

EDUCATION

Bachelor of Mechanical Engineering (Hons.) | **Universiti Malaya (UM), Kuala Lumpur** **2017 – 2021**
First Class, CGPA 3.74/4.00 | Dean's Award, five-time recipient (2017–2021)

Foundation in Physical Science | **Pusat Asasi Sains UM (PASUM)** **2016 – 2017**
CGPA 4.00/4.00

SELECTED PROJECTS

- **Cantilever Pin Introduction, MCT SH-5300 / FH-1200 Handlers (NXP):** Raised contact pin life from 300k to 1M touchdowns, reducing downtime and improving MTBF. Applied to VSON8/14, SO8/14, HVQFN8/18 and DHVQFN18 fine-pitch NiPdAu leadframe packages (incl. automotive CAN/CAN FD transceivers) and integrated into the Cohu Jaguar in-strip handler.
- **NPI High-Speed Ethernet Switch, Final Test Hardware (NXP, 2024):** 5nm 16G automotive Ethernet switch. Delivered new-generation test hardware on a tight budget and timeline with FTR and OTD.
- **MPC8308 J973 to UP1600 Conversion (NXP, 2024):** Moved production to a faster test platform to meet a demand ramp; secured ~US$40M revenue.
- **NPI USB Type-C 48V CC/SBU Protection IC (NXP, 2023):** Delivered final test hardware for the NPI release.
- **Automatic Feeder Stamping Machine (Top Glove, 2022):** Automated stamping and sticker printing with Power BI traceability; RM88,000 annual saving, 2 manpower eliminated.
- **Packer ID Traceability Kiosk (Top Glove, 2022):** Digitalised Packer ID recording in Power BI; RM40,000 annual saving.

CERTIFICATIONS & TRAINING

- **AI & Data Science:** Machine Learning Specialization (DeepLearning.AI & Stanford) (2026); IBM Data Science Professional Certificate (2026); Google AI Professional Certificate (2026)
- **Quality:** Lean Six Sigma Green Belt (2024); NXP Yellow Belt Six Sigma (2022)
- **Professional:** Graduate Engineer, BEM (2022); Graduate Technologist, MBOT (2022)
- **Technical:** Semiconductor Digital Device Test Methodology (2023); RF Test Board Design (2024); PCB Layout Design Techniques (2025); SOLIDWORKS FEA and CFD (2023); Microsoft Excel Advanced (2024)
- **Cloud:** AWS & Microsoft Azure Cloud Practitioner (2023)

AWARDS

- SPOT Award Q3 2025: Innovative risk mitigation on multiple NPI qualifications
- SPOT Award Q3 2024 (two): J973 to UP1600 conversion securing US$40M revenue; NXP Innovation World Tour Kuala Lumpur 2024 (Global Hardware)
- Technical Excellence Award Q3 2023: Bent lead elimination on S32K396 176LQFP product

TECHNICAL SKILLS

Project & Quality: NPI/NTI lifecycle, scheduling, risk management, DOE, DFMEA, MSA, SPC, 3x5 Why, Lean Six Sigma, ISO 9001:2015, IATF 16949

Test Platforms: Teradyne UltraFlexPlus, UltraFlex UP1600, J750; Advantest Exascale Duo, PinScale PS1600; handlers: MCT SH-5300/FH-1200, Delta Matrix/Castle, Esmo Talos, Seiko Epson, Cohu Jaguar

Design & Simulation: Cadence Allegro, SOLIDWORKS (Modelling, FEA, CFD), AutoCAD, DraftSight

PLM & Project Tools: ARAS Innovator, Agile PLM, CollabNet, NXP internal NPI Tools and Internal Collaboration Portal (ICP)

Data & Analysis: Power BI, JMP, Minitab, Advanced Excel (automated Gantt scheduling and project tracking), SQL

AI & Machine Learning: Python (NumPy, Scikit-learn, TensorFlow, Keras), machine learning, data science, applied AI

Languages: Malay (Native), English (Proficient)

References available on request.